

ParkWise Nairobi

A Parking Availability Prediction Platform for Nairobi

Group Capstone Project Proposal

Program: Data Science Fellowship

Date: August 2026

Step 1: Business Understanding

1. Business Overview

Nairobi's parking shortage is well documented. A 2011 IBM commuter survey found that motorists in Nairobi spend an average of 31.7 minutes searching for a parking spot, well above the 19.8 minute global average measured in the same study. A 2019 National Assembly Public Accounts Committee report found that Nairobi, a city with more than 1.3 million registered vehicles, had only around 6,125 formal parking spaces, most of them operated by Nairobi City County. The county has since digitised parking fee payment through platforms such as nairobiservices.go.ke and the Nairobi Pay app, and a growing number of private ventures, Spot Finder, Parksby, Naiparq, and KERB among them, now offer app-based parking discovery and payment across zones such as the CBD, Westlands, Kilimani, and Upper Hill. What none of these platforms currently do, based on their public descriptions, is tell a driver what will be available at a future time, only what is listed, bookable, or reportedly open right now.

This project proposes a data-driven prediction platform that lets a driver check, ahead of time, whether parking is likely to be available at a chosen location for a future date and time, whether that is later the same day or a week ahead, using patterns in historical occupancy, time of day, day of week, and holiday effects, rather than only reporting current state.

2. Problem Statement

Drivers in high-demand Nairobi areas (CBD, Westlands, Kilimani, Upper Hill, and similar zones) cannot currently plan a trip around parking availability in advance, and instead discover whether space exists only by arriving and circling, or by checking an app that reports current occupancy rather than a forecast. This wastes time, fuel, and contributes to congestion in already constrained areas, and it leaves both drivers and facility operators reacting to parking pressure rather than anticipating it.

3. Challenges

- No public, official occupancy dataset exists for Nairobi parking facilities comparable to open sensor datasets available for some other cities, so ground-truth occupancy data has to be collected directly by the project team rather than downloaded.
- Existing Nairobi parking apps that mention real-time availability (Spot Finder, Parksby, KERB) rely on proprietary vendor or partner data that is not publicly accessible for research or model-building.
- Manual field data collection is labour- and time-intensive, and realistically limited to a small sample of locations and time windows rather than citywide, continuous coverage.
- Parking demand is influenced by many overlapping factors at once, time of day, day of week, public holidays, nearby events, and general traffic conditions, which are hard to fully capture with a small primary dataset.

- Any third-party proxy signal considered as a supplement, such as aggregated place-busyness data, reflects general foot traffic rather than verified parking occupancy, and cannot be treated as ground truth.
- A model trained on a small, location-limited sample will not generalise citywide without being explicit about that limitation.

4. Proposed Solution

This project proposes a two-part Parking Availability Intelligence Platform for Nairobi.

The first and primary layer is the Availability Predictor. Using occupancy data collected directly by the project team at a defined sample of parking locations across multiple time windows, combined with day-of-week, time-of-day, and Kenya public holiday features, the model predicts the likelihood that parking will be available at a chosen location for a future date and time selected by the user. This is the core, buildable deliverable and the one the platform depends on.

The second layer, Facility Review Intelligence, is a smaller, optional extension. Where public reviews exist for a surveyed facility, review text is classified into themes (availability, price, safety, attendant service) to help distinguish whether driver frustration at a given location stems from genuine capacity shortage or from a fixable service issue, similar in spirit to the review-theme layer in the reference CRISP-DM plan this proposal is adapted from.

Scope note on deliverable structure: the Availability Predictor (Layer 1) is designed to stand alone as a complete, fully evaluated deliverable — a working prediction model trained and tested on the team's own field-collected data, scoped honestly to the surveyed locations. The Review Intelligence layer (Layer 2) is an optional extension that depends on review volume being sufficient at the surveyed facilities. If review volume proves too thin, Layer 1 alone, fully built and evaluated, is treated as the minimum viable submission.

5. Objectives

- To design and run a structured field data collection process across a defined sample of Nairobi parking locations, capturing occupancy counts, timestamps, and zone information.
- To build a Kenya-specific holiday and day-of-week feature set to capture demand pattern effects.
- To train a model predicting the likelihood of parking availability at a chosen location for a future date and time, evaluated against a naive baseline (e.g. the historical average occupancy for that hour and day type).
- To construct a labeled evaluation set from held-out observation windows, and measure prediction accuracy against it rather than informal spot-checking.
- Where review data is available for surveyed facilities, to classify reviews into operational themes (availability, price, safety, attendant service) to support facility-level diagnosis.
- To design a dashboard where a user selects a location and a future date/time and receives a predicted availability likelihood.
- To clearly document the coverage limits of the field-collected dataset (locations, time window, sample size) so predictions are not presented as citywide or production-grade.

6. Success Criteria

Technical Success

- An occupancy dataset collected across a defined sample of locations and time windows, with documented coverage and no unexplained gaps in the collection schedule.

- A prediction model that meaningfully outperforms a naive “historical average for that hour and day type” baseline on held-out observation windows.
- If Layer 2 is built, a review-theme classifier that meaningfully outperforms a majority-class baseline on held-out labeled reviews.

Business Success

The platform should allow a driver to answer, from the dashboard alone:

- Am I likely to find parking at this location at the time I plan to arrive?
- Which times or days at this location tend to be busiest, so I can plan around them?
- If reviews are available, is congestion at this location a genuine capacity issue or a fixable service issue?

User Success

- Drivers get a forecast to plan around, rather than only a snapshot of current conditions or a listing of bookable spaces.
- The platform is explicit about which locations and time windows are covered by real field-collected data, so confidence in a prediction is not overstated for uncovered areas or times.

7. Target Users

- Drivers who regularly commute to or through high-demand areas (CBD, Westlands, Kilimani, Upper Hill) and want to plan a trip around likely parking availability, whether later the same day or in the days ahead.
- Facility operators and Nairobi City County parking planners, who could use demand pattern findings to inform pricing, staffing, or capacity decisions at surveyed locations.
- Secondly, ride-hailing and delivery drivers, whose efficiency is directly affected by time spent searching for parking or drop-off space in the same high-demand zones.

8. How the User Interacts with the Platform

- A user selects a parking location (from the surveyed sample) and a future date and time, and receives a predicted likelihood of finding available parking, along with the busiest nearby time windows for that location.
- Where review data exists for that location, the user also sees a short summary of the dominant review themes (e.g. frequent complaints about availability versus safety or service).
- A Streamlit-style dashboard prototype is appropriate for demonstrating the platform in this project. A citywide production version would require a data-sharing arrangement with Nairobi City County or an existing private operator (such as those with sensor or transaction-level occupancy data) rather than relying on manually collected field data; this is noted as a scope boundary for the prototype phase.

9. How This Problem Has Been Solved Before

- Naiparq is a peer-to-peer Nairobi parking platform that lets space owners list unused parking and lets drivers locate, reserve, and pay for a listed space, addressing the discovery-and-payment problem but not forecasting future demand at a location.
- Spot Finder advertises real-time available bay counts updated roughly every 30 seconds across CBD, Westlands, Kilimani, Upper Hill, Parklands, Industrial Area, Gigiri, and Karen, with M-Pesa payment and physically inspected listings, but this reflects current state at the moment of checking, not a forecast for a future date or time.

- Parksby operates live in Nairobi (CBD, Westlands, Kilimani, Parklands) and states it uses analytics to provide real-time availability, pricing, and usage trends, but its public description centres on present-moment booking rather than a forward-looking prediction a user could check a week ahead.
- KERB is referenced in local coverage as an app that helps drivers find real-time parking availability in Nairobi CBD, again a current-state discovery tool rather than a predictive one.
- Nairobi City County's own digital channels, nairobiservices.go.ke and the Nairobi Pay app, digitise parking fee payment and permit applications, and represent the county's main investment in parking technology to date, but they are transactional (payment, permits) and do not report or predict availability at all.
- Across all of these, the consistent gap is the same: existing Nairobi parking technology tells a driver what is being listed, bookable, or reportedly open right now. None of them, based on public descriptions, predict what will likely be available at a specific future date and time using historical demand patterns.

10. How This Proposal Is Different

- This project does not attempt to compete with or replace real-time discovery and payment apps like Spot Finder, Parksby, or Naiparq, or the county's payment platforms. It targets a capability none of them currently offer: a forward-looking prediction for a future date and time, not just a present-moment reading.
- The model is trained on data the project team collects and controls, with an honestly documented, limited coverage area, rather than presenting a small sample as citywide, production-grade coverage.
- Where a review-theme layer is included, it connects survey-based demand patterns to the qualitative reasons behind driver frustration at a given location, an integration not offered by any of the discovery or payment apps reviewed above.

11. Decisions This Solution Will Inform

- Drivers deciding when to leave, which location to head to, or whether to use an alternative to driving, based on a predicted likelihood of finding parking rather than guesswork.
- Facility operators or county planners deciding where demand pressure is highest at which times, to inform pricing, staffing, or signage decisions at surveyed locations.
- Where a review-theme layer is included, whether driver complaints at a given facility point to a genuine capacity shortfall versus a fixable service issue.
- Where a future production version would need to invest in a formal data-sharing arrangement with Nairobi City County or an existing operator to move from a limited field sample to full-coverage, continuously updated occupancy data.

12. Data Science Task Definition

- Task 1: availability prediction (Layer 1, primary deliverable). Predict the likelihood of parking availability at a surveyed location for a future date and time, using time-of-day, day-of-week, holiday flag, and historical occupancy pattern features. Framed as classification (available / likely full) or regression (predicted occupancy rate), evaluated against a naive "historical average for that hour and day type" baseline on held-out observation windows.
- Task 2: review theme classification (Layer 2, optional extension). Where sufficient review volume exists for a surveyed facility, classify review text into operational themes (availability, price, safety, attendant service, other). Evaluated against a majority-class baseline on held-out labeled reviews, and only pursued if review volume at surveyed facilities supports it.

- Supporting task: field data collection design and documentation. Because no public Nairobi occupancy dataset exists, the project team must design and execute a structured observation schedule (which locations, which time windows, how occupancy is counted and recorded), and document the resulting dataset's coverage limits as part of the deliverable, not as an afterthought.

Step 2: Data Understanding

Planned Data Sources

Layer 1 (Availability Predictor)

- Primary field-collected occupancy data: manual observation surveys carried out by the project team at a defined sample of parking locations (for example, selected streets/lots in the CBD, Westlands, Kilimani, and Upper Hill), across multiple time windows over several weeks. Each observation records occupied and total spaces, timestamp, and location. This is the project's own ground truth, and its coverage is intentionally scoped to what the team can realistically and consistently observe.
- Nairobi City County parking zone and rate reference data (Zone I to Zone IV classifications and associated rates), used as categorical and pricing context rather than as an occupancy source.
- Kenya public holiday calendar, sourced via the Python `holidays` package (Kenya locale) or an equivalent public holiday list, used to flag holiday and near-holiday effects on demand.
- Optional secondary proxy: aggregated place-busyness data for named locations where publicly viewable, used only as a rough cross-check on time-of-day patterns, not as ground truth, since its underlying methodology is not published or verifiable.

Layer 2 (Review Intelligence, optional)

- Public reviews for named Nairobi parking facilities corresponding to the surveyed locations, collected from public review platforms (e.g. Google Maps) and joined by facility name or address, following the same approach used for the Birmingham reference project this proposal is adapted from.

Data Calibration Approach

Unlike cities with an open, sensor-based occupancy dataset, Nairobi has no public dataset of this kind, so the project's primary occupancy data is collected directly by the team rather than sourced externally. This means the dataset's scale and coverage are inherently smaller and more limited than an automated sensor feed would provide. The project treats this honestly: predictions are scoped to the specific locations and time windows actually observed, and are not presented as citywide or continuously updated. Zone and rate reference data and the Kenya holiday calendar are used to add structured context around the primary field data, not to substitute for occupancy observations themselves.

Expected Columns and Fields — Layer 1 Occupancy Data

Column / Field	Description	Type	Source
Facility ID / Name	Identifier or name of the surveyed parking location	Categorical	Field survey
Zone	Nairobi County parking zone classification (I-IV)	Categorical	County reference data
Observation Date	Date of the occupancy observation	Date	Field survey
Observation Time	Time of the occupancy	Time	Field survey

	observation		
Day of Week	Derived from observation date	Categorical	Derived
Holiday Flag	Whether the observation date is a public holiday	Boolean	Kenya holiday calendar
Occupied Count	Number of occupied spaces observed	Numerical	Field survey
Total Capacity	Total spaces at the location	Numerical	Field survey
Occupancy Rate	Occupied Count ÷ Total Capacity	Numerical	Derived
Collector ID	Identifier for which team member logged the observation, for QA	Categorical	Field survey

Expected Columns and Fields — Layer 2 Review Data

Column / Field	Description	Type	Source
Facility ID / Name	Matches the Layer 1 facility identifier	Categorical	Public review platform
Review Text	Free-text customer review	Text	Public review platform
Rating	Star rating accompanying the review, if available	Numerical	Public review platform
Review Date	Date the review was posted	Date	Public review platform
Theme Label	Hand-labeled theme (availability, price, safety, attendant service, other)	Categorical	Hand-labeled

Planned Data Quality Checks

Check	What We Will Verify
Completeness	Confirm the planned observation schedule (locations × time windows) was actually completed, with no unexplained gaps.
Consistency	Confirm facility names, zone labels, and capacity figures are recorded consistently across all observations of the same location.
Accuracy	Spot-check a sample of logged observations against photo evidence or a second observer's count, where available.
Uniqueness	Check for duplicate observation entries for the same location, date, and time.

Outliers	Flag any occupied count exceeding recorded total capacity, or implausible jumps between adjacent observations.
Shape and Balance	Confirm observations are spread across a reasonable mix of times of day, days of week, and holiday/non-holiday dates, rather than clustered in one narrow window.

Scope Note

This project's occupancy predictions are limited to the specific locations and time windows the team is able to physically survey, since no public, citywide occupancy dataset exists for Nairobi to draw on instead. This is a demonstration of the prediction approach and methodology, calibrated to real field observations rather than invented data, but it is not a validated citywide production system. A production deployment covering all of Nairobi would require a formal data-sharing arrangement with Nairobi City County or an existing private operator with sensor or transaction-level occupancy data (such as Spot Finder or Parksby), rather than relying on manually collected field samples.